POSITION PAPER / OCTOBER 2026

The Aftermarket Imperative

Unlocking the work frontier AI will not do by default

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Paper + bibliography

Research hypothesis. No new benchmark, market estimate or ROI.

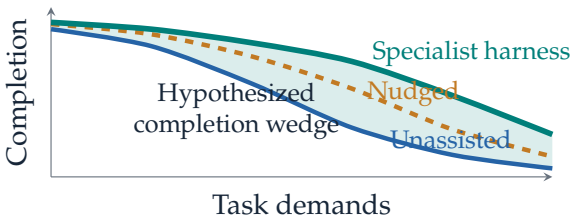
An accepted result is the saleable unit

01 / THESIS

A stock product may leave context, transfers and verification to its user.

A specialist harness supplies a task contract, permitted reach, persistence and proof.

Methods: (Zhou et al., 2023); (Drouin et al., 2024). Historical benchmarks motivate a test; they do not rank this harness.

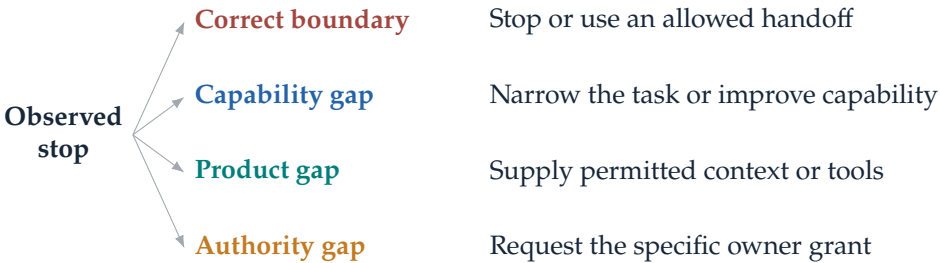


Hypothesized delivery gap. Hand-authored curves; no measured lift.

Claim: a commercial opportunity. Evidence: a position paper. Limit: no comparative result yet.

An observed stop is a diagnosis problem

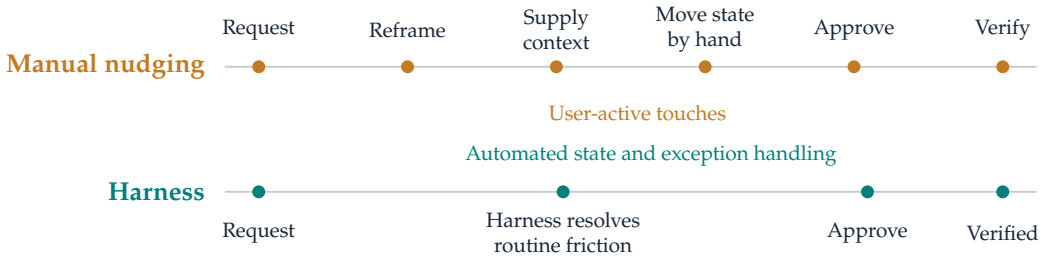
02 / BOUNDARY



Causes may overlap. Refusal is not evidence of hidden capability. Methods: (Rottger et al., 2024); (Cui et al., 2025).

Nudging is hidden labor

03 / USER EFFORT



Retain owner approvals and independent verification on both paths.

Schematic responsibilities, not measured minutes saved. Reframing and context transfer are human work.

Measure user-active minutes, reviewer effort and repair work alongside accepted outcomes.

A harness is a translation contract

04 / MECHANISM

A translation contract with the base model held fixed		
01	**Specify**	Set an observable goal and a permitted route
02	**Reach**	Bind context, tools, accounts and checkpoints
03	**Persist**	Carry state and exceptions across turns and waits
04	**Prove**	Verify the external result and count human effort

Conceptual obligations, not four shipped capabilities. Each implementation needs its own artifact, scope and acceptance evidence.

Where a specialist might win

05 / MARKET ENTRY

Look for

- Recurring, valuable task families
- Fragmented permitted context
- Observable acceptance states
- Exceptions worth productizing

Be skeptical when

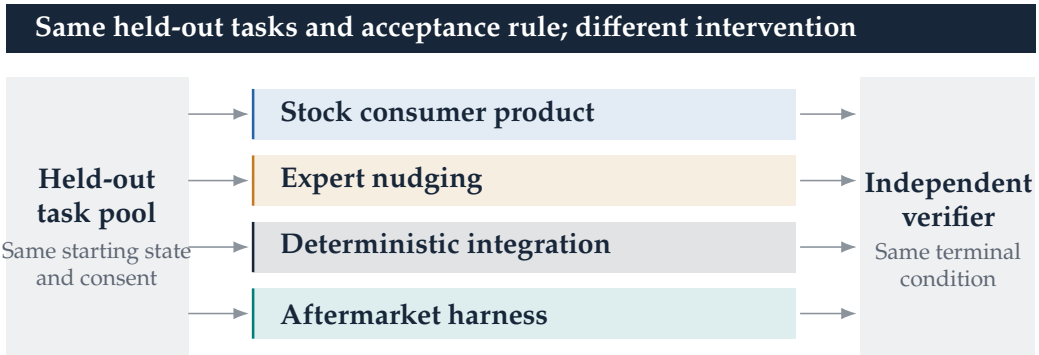
- A stable API solves it directly
- Human judgment is the core work
- Access or permission is absent
- Support consumes the margin

Credentials, enrollment and authorized defensive work illustrate workflow friction; they do not establish a market.

The commercial opportunity is a hypothesis to test against conventional software and human work.

Run the comparison that can fail the thesis

06 / EXPERIMENT



Proposed experiment, not an executed ranking. Match stock/harness model versions. Methods: (Yao et al., 2024); (Kapoor et al., 2024).

Count accepted outcomes, not fluent attempts

07 / ECONOMICS

$$\textbf{Cost per accepted task} = \frac{\text{model} + \text{tools} + \text{human time} + \text{support} + \text{maintenance}}{\text{independently accepted completions}}$$

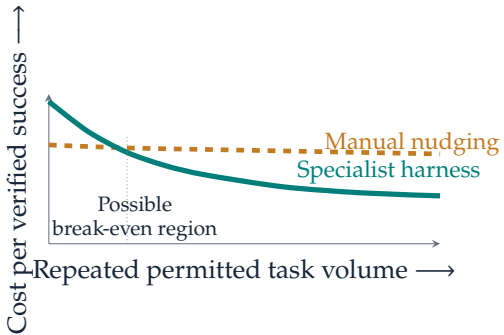
Outcome	Counting rule
Accepted	Count only independently verified completions
Pending / unknown	Report separately; reconcile before retry
Correctly stopped	Report as a boundary, not a completion

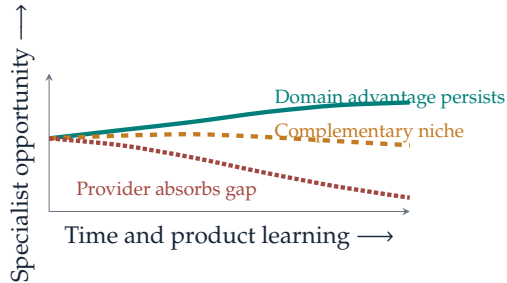
No accepted completions: unit cost is undefined, not zero. Report actual spend separately from elapsed time and disclose labor rates.

Claim: delivery can create value. Limit: a technical win can still lose commercially.

The wedge has a moving boundary

08 / DURABILITY





Conceptual scenarios, not trajectories or forecasts.

Conceptual amortization; no measured break-even point or ROI.

Retest after provider upgrades: specialist value may persist, complement the platform, or disappear.

The verdict must be earned

09 / RESEARCH AGENDA

Test recovery, human effort and component effects with the model held fixed. Retest on held-out services and after upgrades.

Artifact	Evidence and limitation
Paper + bibliography	Position paper; bibliography retained; no new benchmark
Editable source	Buildable TeX; source is not deployment evidence
Synthetic refinery demo	Synthetic UI and simulated evaluation
Pinned test source ecd6c3e	Authored assertions; not rerun in this paper
Dated repository index	Navigation, not proof that each product works

See (Kapoor et al., 2024). No adoption, production, career, market-size or ROI claim.